

Housing over three stages of life

A test of the proposed change

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Adding a mature working period can generate the intended housing comparison with an 80% mortgage. It also permits analytical household solutions. It does not, by itself, deliver a simple formula for equilibrium prices or a fertility result along a policy transition. This calculation tests the change; it does not replace the model in the seminar deck.

1. What changes. A household is young, mature, then old. Its incomes are y^y, y^m, y^o , and its initial financial endowment is b ; write $w = y^y + b$. Income and wealth may differ across households. Fertility n is chosen when young. Children require goods and space in both family stages:

$$\begin{aligned} u^y &= \log(c^y - \chi n) + \alpha \log(h^y - \kappa n) + \vartheta \log n, \\ u^m &= \log(c^m - \chi n) + \alpha \log(h^m - \kappa n), \\ u^o &= \log c^o + \gamma \log h^o + \omega_B \log e. \end{aligned}$$

Lifetime utility is $u^y + \beta u^m + \beta^2 u^o$, plus the existing tenure taste. Here e is the estate, including the retained home's liquidation value. Tenure is retained, homes can be resized, and each home's size is bounded by H_R or H_O , with $H_R < H_O$.

At a stationary price P , let q be the price of a bond paying one unit next stage, τ the property-tax rate, and T the common rebate. The current cost of housing services is $p = DP$, where $D = 1 - q + q\tau$. Net financial assets entering maturity and old age are a_m and a_o . An owner's budgets are

$$\begin{aligned} c^y + qa_m + (1 + q\tau)Ph^y &= w + T, & qa_m + \phi Ph^y &\geq 0, \\ c^m + qa_o + (1 + q\tau)Ph^m &= a_m + Ph^y + y^m + T, & a_o &\geq 0, \\ c^o + qe + ph^o &= a_o + Ph^m + y^o + T, & e &\geq Ph^o. \end{aligned}$$

A renter pays ph at each age, has no housing-sale receipts, and faces $a_m, a_o \geq 0$. Its old budget is $c^o + qe + ph^o = a_o + y^o + T$. The original logistic tenure choice and the common rebate are retained.

The economic difference is the repayment date. When the young mortgage binds, resources on entry to maturity are

$$y^m + T + (1 - \phi/q)Ph^y.$$

The entire mortgage is repaid, including interest. Mature earnings can cover that repayment before retirement. The tested restriction $a_o \geq 0$ forbids negative net financial wealth at retirement; it is an additional institutional choice. Where the derived a_o is strictly positive, permitting mature refinancing leaves the household optimum unchanged.

2. How much can be solved explicitly?

Consider an owner whose young mortgage binds, who saves when mature, and whose housing caps and old estate floor are slack. Let $x = c^y - \chi n$ and $s = h^y - \kappa n$ be young consumption and space after child needs. Define

$$\begin{aligned} W &= w + T, & U &= y^m + qy^o + (1 + q)T, \\ L &= (1 - \phi + q\tau)P, & \delta &= (1 - \phi/q)P, & t &= \chi + \kappa p, & t_L &= \chi + \kappa L, & d &= t - \kappa\delta, \\ K &= 1 + \gamma + \omega_B, & G &= 1 + \alpha + \beta K. \end{aligned}$$

Let $x^m = c^m - \chi n$ and $s^m = h^m - \kappa n$ denote mature adult consumption and space. Mature resources after child needs are $B = U + \delta s - dn$. The continuation choices are

$$x^m = \frac{B}{G}, \quad s^m = \frac{\alpha x^m}{p}, \quad c^o = \frac{\beta x^m}{q}, \quad h^o = \frac{\gamma \beta x^m}{qp}, \quad e = \frac{\omega_B \beta x^m}{q^2}.$$

Fixed fertility. Set $A_y = W - t_L n$ and $A_m = U - dn$. Young space solves a strictly concave, one-variable problem:

$$\max_s \log(A_y - Ls) + \alpha \log s + \beta G \log(A_m + \delta s).$$

Its first-order condition is the quadratic

$$\boxed{0 = \alpha A_y A_m + [\delta A_y (\alpha + \beta G) - (1 + \alpha) L A_m] s - \delta L (1 + \alpha + \beta G) s^2.}$$

Select the root with $s > 0$, $A_y - Ls > 0$, and $A_m + \delta s > 0$. This holds at general ϕ , including $\phi = .8$.

Chosen fertility. Put $r = \beta x / x^m$ and $E = 1 + \alpha + \vartheta$. Then

$$s = \frac{\alpha x}{L - \delta r}, \quad n = \frac{\vartheta x}{t_L + dr}, \quad x = \frac{W + Ur}{E + \beta G},$$

where r solves

$$\left(1 + \frac{U}{W}r\right) \left(1 + \frac{\alpha L}{L - \delta r} + \frac{\vartheta t_L}{t_L + dr}\right) = E + \beta G.$$

Clearing denominators gives a cubic. Its economically relevant root is unique. The income ratio U/W is strictly decreasing in that root: higher mature resources relative to young cash lower r . Thus a condition $r < r_*$ can be written as an explicit income-ratio inequality without solving the cubic.

These formulas must satisfy $0 < r < q$, the housing limits, and

$$a_o = \left(\frac{\beta K}{q} - \frac{\alpha}{D}\right) x^m - (y^o + T) - \kappa P n > 0, \quad \omega_B D > q\gamma.$$

In particular, $\beta K D > \alpha q$ is necessary for this saving regime. Higher mature earnings alone cannot repair a failure of that inequality.

3. The planner and fertility comparisons remain short.

At the stationary reference, match the common distribution of types and tenure across the three current ages. Bars denote means under this distribution. Hold fertility, future real opportunities and old estates fixed. The planner chooses all current consumption and housing, giving equal weight to the remaining utility of each current household. It respects both current resource constraints and each household's tenure cap, and may relax private finance limits.

After subtracting the fixed child needs, total current goods are $\bar{x}^y + \bar{x}^m + \bar{c}^o$. Hence every household receives adult consumption

$$X = \frac{\bar{x}^y + \bar{x}^m + \bar{c}^o}{3}.$$

For a type with fertility n_i and cap H_i , planner housing is

$$h_i^{y,F} = h_i^{m,F} = \min\{H_i, \kappa n_i + z\}, \quad h_i^{o,F} = \min\{H_i, (\gamma/\alpha)z\},$$

where z clears the housing stock. Title and financial-claim adjustments keep future entering resources and net estates fixed. The required current transfers sum to zero because current goods and housing totals are unchanged.

When planner caps are slack, the young receive more housing exactly when

$$\alpha(\bar{s}^m + \bar{h}^o) > (\alpha + \gamma)\bar{s}^y.$$

At $\alpha = \gamma$, mature and old adult space together must exceed twice young adult space. The mature cohort is part of the comparison.

A particularly simple sufficient statement also allows binding caps. If $\alpha \geq \gamma$, fertility is positive and some total capacity is unused, the planner gives each family age more than one third of the housing stock. Therefore, if the competitive young hold at most one third, their aggregate housing rises. This is a utilitarian allocation comparison, not a Pareto claim.

Private fertility after a current allocation change. Children still require goods and housing when mature. The private condition is

$$\frac{\vartheta}{n_i} = \frac{\chi}{x_i^y} + \frac{\alpha\kappa}{s_i^y} + \beta \left(\frac{\chi}{x_i^m} + \frac{\alpha\kappa}{s_i^m} \right).$$

Give a parent its new current consumption and housing bundle, while preserving its continuation problem. At the original fertility, the future marginal cost is the same in both comparisons. Strict concavity then gives

$$n_i^{\text{new}} > n_i \iff \chi \left(\frac{1}{x_i^y} - \frac{1}{X} \right) + \alpha\kappa \left(\frac{1}{s_i^y} - \frac{1}{s_i^{y,F}} \right) > 0.$$

Thus more housing and weakly more adult consumption increase fertility. A housing gain alone need not do so. This is a household comparison at the same continuation opportunities; a market policy changes prices and future choices as well, and needs its own argument.

4. What the equilibrium check establishes.

There is an analytical stationary-equilibrium construction with arbitrary fixed $\phi \in (0, 1)$, including .8. Young and old endowments have bounded heterogeneous support. Mature earnings are sufficiently high relative to young cash resources. Take $\alpha \geq \gamma$. At zero tax, also require

$$\omega_B(1 - q) > q\gamma, \quad \beta(1 + \gamma + \omega_B)(1 - q) > q\alpha.$$

Finite inequalities bound the required mature earnings, rental capacity and owner capacity. On an explicit positive price interval they imply: young owners borrow maximally; mature owners save; both later owner homes exceed the young home; and mature and old renters occupy their rental cap while young renters remain below it. The cap still affects young fertility through the family's future mature housing needs.

Both tenure-specific fertility choices exceed replacement at the lower price and fall below it at the upper price. These inequalities hold type by type, so endogenous logistic tenure shares preserve them after aggregation. Continuity gives a stationary equilibrium inside that interval. At such an equilibrium the young hold less than one third of housing. The fixed-fertility planner described on page 3 gives them more than one third.

This is an analytical existence proof with finite primitive bounds, not a numerical equilibrium followed by a neighbourhood argument. It is nevertheless a deliberately strong sufficient construction: the earnings bounds can be large, and it does not establish the ordering in every possible equilibrium outside the constructed price interval.

Further finite bounds on mature earnings and rental capacity make every young household receive both more consumption and more housing under the dated planner. The private fertility test on page 3 is then strictly positive for every young parent. This establishes a conditional fertility response at unchanged continuation opportunities, not a future market-clearing allocation.

Taxes. The strict construction extends to sufficiently small positive rebated property taxes. That extension uses continuity and a two-variable fixed-point argument; the maximal admissible tax rate has not been derived. Independently, every stationary equilibrium obeys the explicit rebate bound

$$T \leq \frac{\tau(\bar{w} + q\bar{y}^m + q^2\bar{y}^o)}{(1 - q)[3q - \tau(1 + 2q)]}, \quad \tau < \frac{3q}{1 + 2q}.$$

This bound alone does not establish the desired household regime at a chosen tax.

A less extreme household check. With $q = .5$, $\tau = .1$, $\phi = .8$, $\alpha = \gamma = \vartheta = \kappa = 1$, $\omega_B = 2$, and $\beta \in [.49, .5]$, an explicit conditional income interval delivers binding young credit, mature saving and larger old homes with mature resources between 2.5 and 2.8 times young resources. This check also requires $\chi/(\kappa p) \geq 10$ and comparable young and old cash resources. It verifies household feasibility; it is not the stationary construction above.

Assessment. The extra earning stage can generate the intended housing mechanism with conventional origination finance. Fixed-fertility household choices are quadratic; endogenous fertility is cubic. With heterogeneous types and tenure choice, stationary prices still solve aggregate clearing equations, and have no demonstrated closed-form solution. A short general primitive condition covering housing, fertility and a policy transition has not been obtained. The model change therefore merits consideration, but this calculation does not justify replacing the seminar theory wholesale.

Full budgets, derivative checks and the finite equilibrium inequalities are retained together in the existing simplified-OLG evidence folder, in `three_stage_tractability_evidence.md`. No numerical model runs were used.